

Real-Time Construction Site Object Detection for Autonomous Vehicles

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ABSTRACT

This study evaluates the effectiveness of a YOLO-based object detection system for identifying construction-zone elements in autonomous-driving scenarios. Because existing autonomous-driving datasets lack construction-specific imagery, 1,000 dashcam images were manually extracted and annotated from the nuScenes dataset to create a dedicated construction-site dataset. A YOLOv11s model was trained on six object classes using an 80/20 split and standard augmentation procedures. The model achieved an overall mAP@0.5 of 0.779 and mAP@0.5–0.95 of 0.461, with highest performance on visually distinctive, frequently occurring classes such as cones, construction cylinders, and signs. Lower performance on fences and construction vehicles resulted from the lack of training instances and class imbalance. Barriers suffered from segmented bounding box annotations, leading to overestimation. Inference speed operated well within real-time requirements, with a latency of ~ 7 ms per frame. Results demonstrate that lightweight YOLO architectures can support construction-zone perception without additional sensors, while highlighting the need for improved dataset balance and segmentation-based annotations to enhance robustness in visually complex environments.

I. INTRODUCTION

Autonomous systems in modern vehicles have progressed rapidly, pioneered by Advanced Driver Assistance Systems (ADAS) such as Active Cruise Control, Lane Keep Assist Systems, and Autonomous Emergency Braking. Recently, for full autonomous driving systems, camera-based perception has emerged as a cost-effective alternative to LiDAR and radar; for example, Tesla's supervised Full Self Driving relies solely on cameras to interpret and respond to dynamic road conditions [1]. These developments create a need to evaluate whether modern real-time detectors can reliably interpret high-variability scenes such as roadway construction zones.

As a result, the role of Computer Vision (CV) and object detection algorithms in vehicular contexts is of utmost importance. Among the object detection frameworks developed in recent years, You Only Look Once (YOLO) has emerged as a dominant real-time object detection algorithm. Unlike two-stage detectors such as R-CNN and Faster-RCNN, which first divide an image into region proposals and then classify each individually, YOLO applies a single neural network to the

entire image in one pass, enabling faster results for real-time detection [2].

Despite the staggering improvement in the ability of self-driving vehicles to navigate complex environments, construction sites have continuously proved to be an obstacle. Construction sites are highly variable environments that can contain cones, construction cylinders, barriers, and other objects with significant differences in layout and visual appearance. Therefore, it is imperative that vision-based object detection algorithms can accurately discern elements within a construction site environment.

Previous research has determined that construction sites are areas of vulnerability for autonomous vehicles and has implemented varying methods to find a solution. One study developed a computer vision model on simulated data in the Car Learning to Act (CARLA) simulator. YOLOv8 was used to detect construction zone objects similar to this study; however, the training data was not based on any real-life footage. Despite this limitation, the authors were still able to train on a larger dataset due to the ease of access and augmentation of the data [3].

Another study used the YOLO object detection algorithm to identify the presence of entire construction zones in dashcam footage and used GPS data for localization. The study used images from Boston and Singapore derived from the nuScenes dataset and included custom images from Helsinki. [4].

Both studies compared multiple object detection models such as Retinanet, R-CNN, and Fast R-CNN, alongside YOLO. The results collected in both studies showed the potential and effectiveness of the YOLO framework.

The objective of this paper is to evaluate the efficacy of a YOLO-based object detection system for identifying construction zone elements in autonomous vehicle navigation contexts. This study uses YOLOv11s, the latest lightweight Ultralytics architecture trained on annotated dashcam images of construction sites.

II. METHODOLOGY

A. Custom Dataset Creation

Several automotive dashcam datasets exist for autonomous driving, but none focus specifically on roadway construction zones; instead, they emphasize elements like lane markings, traffic signs, and pedestrians. Manually collecting dedicated construction-zone footage would be labor-intensive and costly.

Karthikeyan [4] faced a similar issue, ultimately using existing datasets that sparsely contained images of construction sites.

The nuScenes dataset was chosen for this work because it contains approximately 1.4 million visible-light images captured from six synchronized vehicle-mounted cameras, captured under varied weather conditions. It includes data from Boston and Singapore, which differ in lighting, construction objects, and urban color schemes.



Fig. 1. Construction in Singapore



Fig. 2. Construction in Boston

Because nuScenes and other publicly available datasets lack a construction-specific subset, a large portion of the nuScenes was manually reviewed to identify images containing roadway construction elements. The custom dataset used in this study included 1,000 images from the forward-facing camera only, representative of an onboard dashcam.

To annotate the images, the Computer Vision Annotation Tool (CVAT), created by the OpenCV group, was used. The tool includes multiple types of annotation methods, such as bounding boxes, segmentation, polygons, etc. The tool can be used on a browser by uploading a dataset and specifying classes [5]. This study initially began with 7 classes: Cones, Construction Cylinders, Signs, Fences, Barriers, Vehicles (cranes), and Humans (construction workers). Later, the Humans class was omitted to reduce false positives with pedestrians, which posed a risk to model accuracy; however, with a larger dataset and many training examples, construction workers should be included. The classes were annotated with 2D bounding boxes. With CVAT, the annotations can be exported in multiple formats; however, for the purpose of this study, the annotations were exported in the YOLO v1.1 format.

B. Preliminary Auto-Annotation Model

To reduce manual labor, the initial 500 images were manually annotated and used to train a preliminary YOLOv11s model, which was used to annotate the remaining 500 images automatically. Despite the significant improvement in workload efficiency, the auto-generated annotations still required manual review frame by frame in CVAT to correct inaccuracies.

C. Training Procedure

The final 1,000 image dataset was divided into an 80/20 training-validation split. All images and corresponding annotation files were restructured into the directory hierarchy required by the Ultralytics YOLO framework. A custom “data.yaml” file was created to define class names and dataset directory paths. The model was trained using the most recent YOLOv11s. Training was performed on a desktop using an NVIDIA RTX 2080 SUPER GPU. The hyperparameters included a batch size of 4, an image resolution of 1024x1024, 60 epochs, stochastic gradient descent (SGD) optimization, and standard YOLO data augmentation operations such as mosaic and horizontal flipping. A high input resolution was chosen because many objects appeared at a distance and were $\leq 20 \times 20$ px. Consequently, a low batch size was chosen to reduce memory load.

III. RESULTS

The dataset contained 6,433 labeled instances and showed strong class imbalance as cones and barriers were the most frequent classes in construction areas, unlike signs, vehicles, and fences, as shown in Table I.

TABLE I
INSTANCE COUNT PER OBJECT CLASS

Class	Instances
Cone	3519
Construction_Cylinder	268
Signs	593
Fences	135
Barriers	1629
Vehicles	289

After completion of training, the model auto-generated training and validation loss curves for bounding box regression, classification loss, and distribution focal loss are presented in Figure 3. Validation losses showed smaller but consistent decreases during the same period. The diminishing loss values indicate an increase in the model’s ability to accurately predict and classify objects.

Performance was evaluated using **Precision**, **Recall**, **F1 score**, **mAP@0.5**, and **mAP@0.5:0.95** and organized in Table II:

- **Precision** = $TP / (TP + FP)$. High precision indicates few false positives.
- **Recall** = $TP / (TP + FN)$. High recall indicates few false negatives.
- **F1 score** = harmonic mean of precision and recall.

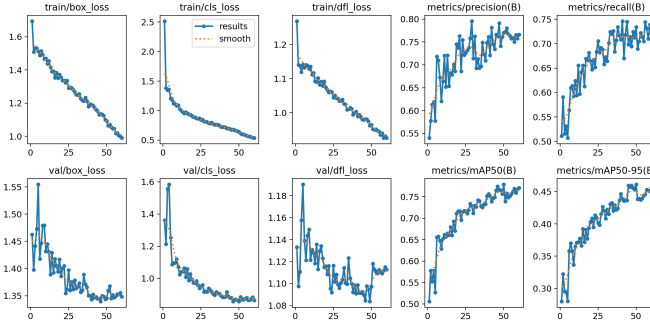


Fig. 3. Performance Curves of the Developed Model

- **mAP@0.5** = mean Average Precision at IoU threshold 0.5.
- **mAP@0.5:0.95** = averaged over IoU thresholds 0.5–0.95 in 0.05 increments, providing a stricter overall accuracy measure.

TABLE II
CLASS-WISE PERFORMANCE METRICS ON VALIDATION SET

Class	Precision	Recall	F1	mAP	
				@0.5	@0.5–0.95
Cone	0.862	0.839	0.850	0.895	0.505
Construction_Cylinder	0.827	0.797	0.811	0.862	0.508
Signs	0.861	0.866	0.863	0.881	0.595
Fences	0.687	0.564	0.620	0.611	0.336
Barriers	0.664	0.651	0.657	0.703	0.440
Vehicles	0.723	0.673	0.697	0.722	0.383
Overall	0.771	0.731	0.751	0.779	0.461

The effect of confidence threshold on F1-score, precision, and recall is shown in Figures 4–6. For all classes combined, the maximum F1-score was **0.75**, occurring at a confidence threshold of **0.343**. At this threshold, individual class F1 values ranged from approximately **0.62** (Fences) to **0.86** (Signs), with Cone and Construction Cylinder achieving F1-scores of approximately **0.85** and **0.81**, respectively.

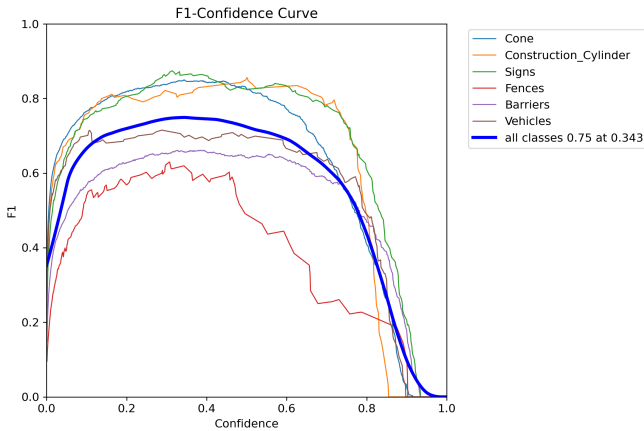


Fig. 4. F1-Confidence Curve

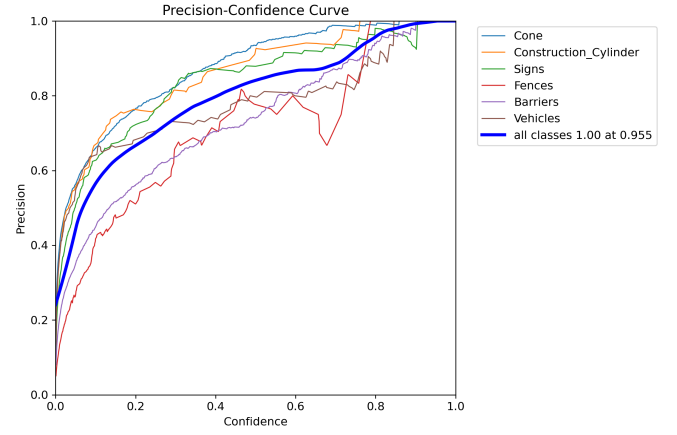


Fig. 5. Precision-Confidence Curve

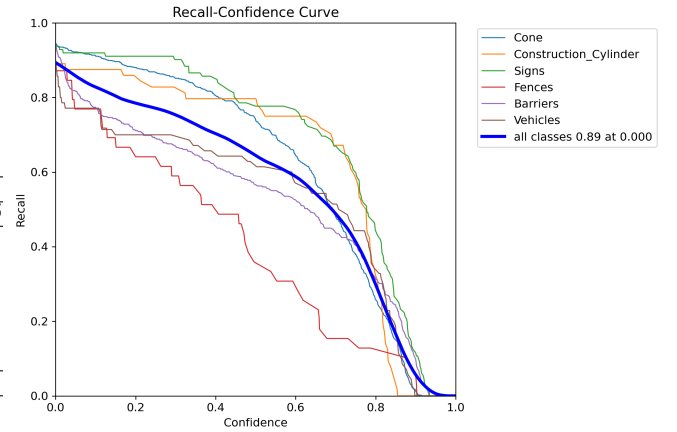


Fig. 6. Recall-Confidence Curve

Precision increased monotonically with confidence threshold and reached **1.00** at a threshold of approximately **0.955** for all classes combined (Figure 5). In contrast, recall decreased as the threshold increased, starting near **0.89** at a threshold of 0.0 and approaching 0 at the highest thresholds (Figure 6).

Figure 7 shows the confusion matrix computed on the validation set at the selected confidence threshold. For the Cone class, **612** instances were correctly classified as cones, while **78** cone instances were misclassified as background. Construction Cylinders showed **53** correct detections, with **10** instances misclassified as background. Signs achieved **102** correct predictions, while **10** sign instances were predicted as background.

Barriers exhibited **283** correct detections, with **108** barrier instances predicted as background and **167** background regions misclassified as barriers. Vehicles recorded **50** correct detections, with **20** vehicle instances predicted as background. Fences showed **25** correct detections and **12** fence instances predicted as background, reflecting the higher difficulty of classes with low instance counts.

The normalized confusion matrix in Figure 8 summarizes

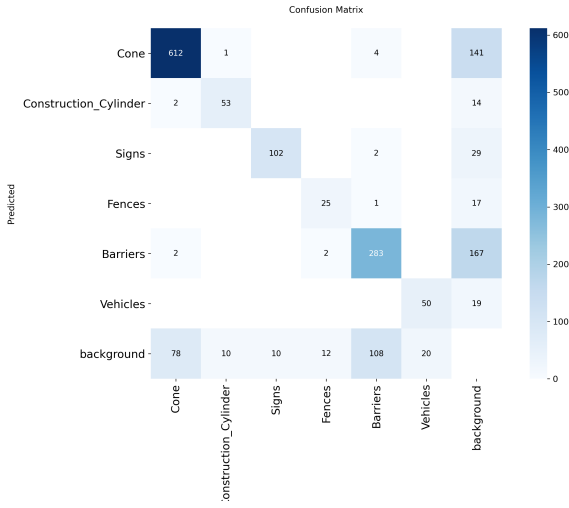


Fig. 7. Confusion Matrix

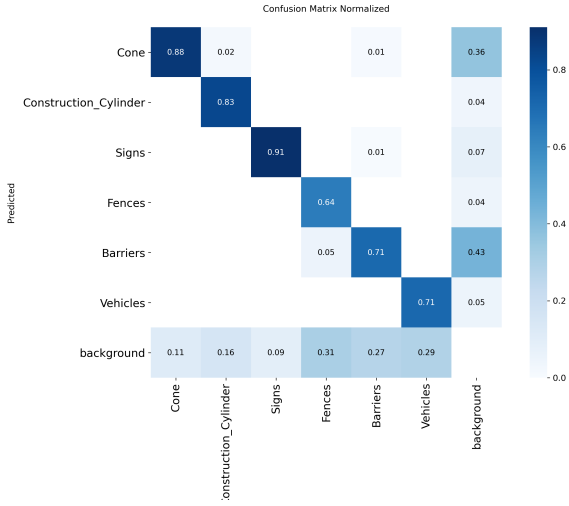


Fig. 8. Confusion Matrix Normalized

the per-class recall values. The diagonal entries show high recall for Signs (0.91), Cone (0.88), and Construction Cylinder (0.83). Barriers and Vehicles achieve moderate recall values of approximately 0.71, while Fences exhibit a lower recall of 0.64. These results indicate strong performance for classes with distinctive geometry and sufficient training samples, and comparatively weaker performance for classes with greater visual ambiguity or fewer training instances

IV. DISCUSSION

The results indicate that YOLO11s successfully detected four of the six classes with moderate to high accuracy. The strongest overall performance occurred for the Cone, Construction Cylinder, and Signs classes, shown by the P, R, and mAP values shown in Table II. The performance of these classes can be correlated to their distinct color, shape, and position.

Classes with fewer instances and greater visual variability, such as Fences and Vehicles, produced lower detection performance. Despite their relatively distinct appearance, the Fences class contained only 135 instances, resulting in the lowest recall and the lowest mAP values due to limited exposure to varying visual circumstances. Similarly, the Vehicles class suffered from a small number of training instances combined with high variability in size, shape, and orientation, which further hindered reliable detection

A particular source of error arose in the Barriers class, as the limitations that explain the poor performance in low-instance classes, such as Fences, do not apply here. Likewise, the class does not suffer from the visual variability seen in the Vehicles class. Barriers are uniform in shape and appear frequently in the dataset, with 1,629 annotated instances. The low precision and recall scores are therefore attributed to the segmented bounding boxes used during annotation (see Figure 9). These boxes were intentionally drawn in smaller sections (usually 3) to reduce the amount of background inclusion, but this approach likely introduced label inflation and caused the model to overestimate the amount of Barrier classes and detect multiple Barrier instances within another (see Figure 10).



Fig. 9. Segmented Bounding Boxes



Fig. 10. Barrier Overestimation

The confusion matrices (Figures 7–8) indicate that the majority of false positives arise from confusion with background regions or visually similar classes. Misclassification of true objects as background is evident across all classes, suggesting that low-contrast instances and partial occlusions reduce discriminability. Signs and Barriers exhibit relatively high normalized diagonal values (0.91 and 0.71, respectively), indicating consistent recognition. In contrast, Fences demonstrate a lower diagonal recall value (0.64) with misclassifications dis-

tributed across background and barrier predictions, supporting the need for additional training samples and improved feature representation for this class.

The training curves in Figure 3 show stable convergence for all loss terms, indicating that 60 epochs were sufficient to avoid underfitting. The absence of late-epoch instability suggests that the model benefited from the selected hyperparameters and that premature overfitting did not occur. However, the plateauing of the mAP values before epoch 60 suggests that additional training beyond this point did not yield meaningful performance gains and that the optimal weights were likely reached earlier in the training process.

The Precision-Recall curves demonstrate that the model maintained relatively high precision across most recall levels for the three highest performing classes (Figure 11). Fences and Vehicles experienced earlier degradation, consistent with their lack of dataset representation and variability within the class.

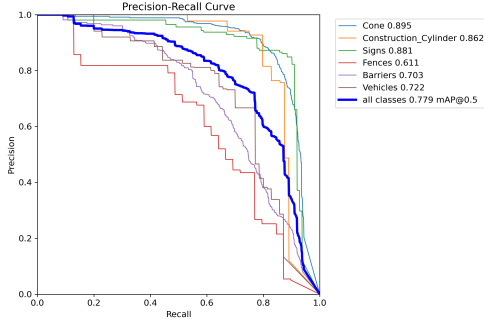


Fig. 11. Precision-Recall Curve

The model's inference speed averaged approximately 5.2 ms per image, with an additional 0.7 ms for preprocessing and 1.2 ms for postprocessing, resulting in a total latency of 7 ms per frame. This corresponds to an effective throughput of more than 140 frames per second, which exceeds the real-time requirements of most autonomous-vehicle perception pipelines, typically benchmarked at 30 frames per second [6]. The processing time, therefore, indicates that the model is computationally efficient and suitable for deployment in real-time environments, assuming comparable hardware and stable input resolution.

Overall, these results are consistent with established literature that highlights the dependence of performance on data set balance, object size, and object frequency. These findings also match documented YOLO behavior in small-object and imbalanced-class scenarios, suggesting that the model's performance aligns with expected limitations under constrained training conditions.

V. CONCLUSION

This study developed a construction-site object detection model using YOLO11s and demonstrated that a lightweight detector can identify key roadside objects with practical accuracy. The model reached an overall mAP@0.5 of 0.779

and an mAP@0.5–0.95 of 0.461. Performance was highest for classes with distinctive visual features and sufficient training instances, underscoring the role of dataset balance and visual variability in supervised learning.

The results indicate that real-time detection in construction environments is feasible with compact architectures. Future work should incorporate segmentation-based annotation, as segmentation would reduce errors caused by occlusion and fragmented bounding boxes (Figure 12). YOLO-segmentation models offered by Ultralytics are well-suited for this task and would address the limitations observed with the Barriers class. Expanding the dataset to include more geographic regions and a broader range of construction scenarios would further improve model generalization and robustness.

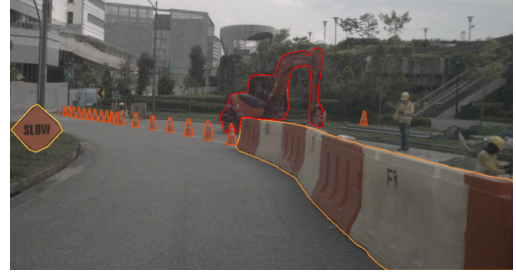


Fig. 12. Example Segmentation Annotation

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